

Fuzzy Logic in Behavioral Finance

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Fuzzy Logic Connectives: Theory and Applications

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The Representative Heuristic

What is it?

The representativeness heuristic is a mental shortcut that we use when deciding whether an object belongs to a class. Specifically, we tend to overemphasize the similarity or difference between the object and class to help us make this decision.

Why it happens?

Our perception of people, animals, and objects relies heavily on categorization: grouping similar things together. Within each category exists a prototype: the “average” member that best represents the category as a whole. When we use the representativeness heuristic, we compare something to our category prototype, and if they are similar, we instinctively believe there must be a connection.

The Representative Heuristic

Example 1 – Representativeness and stomach ulcers

When an Australian doctor discovered that a bacterium, and not stress, causes stomach ulcers, other medical professionals initially didn't believe him because ulcers feel so similar to stress. In other words, stress is a more representative cause of an ulcer than bacteria are.

Example 2 – Representativeness and astrology

The personality types associated with each star sign in astrology are chosen because they are representative of the animal or symbol of that sign.

The Anchoring Heuristic

What is it?

The anchoring bias is a pervasive cognitive bias that causes us to rely too heavily on information that we receive early in the decision-making process. Because we use this “anchoring” information as a point of reference, our perception of the situation can become skewed.

Why it happens?

There are two dominant theories behind the anchoring bias. The first one, the anchor-and-adjust hypothesis, says that when we make decisions under uncertainty, we start by assigning some initial value and subsequently adjusting it, but our adjustments are usually insufficient. The second one, the selective accessibility theory, says that the anchoring bias happens because we are primed to recall and notice anchor-consistent information.

The Anchoring Heuristic

Example 1 – Anchors in the courtroom

In criminal court cases, prosecutors often demand a certain sentence for the accused. Research shows these demands can become anchors that bias the judge's decision-making.

Example 2 – Anchoring and portion sizes

The common tendency to eat more when faced with a larger portion might be explained by anchoring. In one study, participants' estimates of how much they would eat were influenced by an anchoring portion size (large or small) they had been told to imagine previously.

Fuzzy C-Means

Setup

- FCM is a clustering algorithm where data points can belong to multiple clusters with varying degrees of membership.

Variables

- ▶ $X = \{x_1, x_2, \dots, x_m\}$: Dataset with m data points.
- ▶ c : Number of clusters.
- ▶ c_i : Center of cluster i , for $i = 1, 2, \dots, c$.
- ▶ $u_i(x_j)$: Membership degree of x_j in cluster i , where:
 - ▶ $0 \leq u_i(x_j) \leq 1$.
 - ▶ $\sum_{i=1}^c u_i(x_j) = 1$ for all $j = 1, 2, \dots, m$.
- ▶ $m_f > 1$: Fuzzification parameter that controls the degree of fuzziness in the clustering.
- ▶ d_{ij} : Distance between data point x_j and cluster center c_i .

Fuzzy C-Means

Loss Function (Objective Function)

$$J = \sum_{j=1}^m \sum_{i=1}^c u_i(x_j)^{m_f} \cdot d_{ij}^2$$

$d_{ij}^2 = \|x_j - c_i\|^2$: Squared l2 distance

Optimal Values

1. Cluster Centers (c_i):

$$c_i = \frac{\sum_{j=1}^m u_i(x_j)^{m_f} \cdot x_j}{\sum_{j=1}^m u_i(x_j)^{m_f}}$$

2. Membership Degrees ($u_i(x_j)$):

$$u_i(x_j) = \frac{1}{\sum_{k=1}^c \left(\frac{\|x_j - c_i\|}{\|x_j - c_k\|} \right)^{\frac{2}{m_f - 1}}}$$

Algorithm 1: Fuzzy C-Means

Initialize: The membership matrix U such that:

$$\sum_{i=1}^c u_i(x_j) = 1, \quad u_i(x_j) \geq 0, \quad \forall j = 1, 2, \dots, m.$$

Calculate: Cluster centers c_i using:

$$c_i = \frac{\sum_{j=1}^m u_i(x_j)^m \cdot x_j}{\sum_{j=1}^m u_i(x_j)^m}.$$

Update: The membership matrix using:

$$u_i(x_j) = \frac{1}{\sum_{k=1}^c \left(\frac{\|x_j - c_i\|}{\|x_j - c_k\|} \right)^{\frac{2}{m-1}}}.$$

Repeat: Steps 2 and 3 until the objective function J stabilizes (stops decreasing).

$$M = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 2 & 3 \\ 4 & 1 \\ 4 & 3 \\ 5 & 2 \end{bmatrix}$$

Figure 1: Pattern matrix with 6 objects and 2 features

Data Matrix			I. MEMBERSHIP Degrees		Distance	
			Cluster 1	Cluster 2	Between the Object and Center Vector of Cluster 1	Between the Object and Center Vector of Cluster 2
X ₁	1	2	0.9630	0.0370	0.6552	3.3448
X ₂	2	1	0.8532	0.1468	1.0574	2.5493
X ₃	2	3	0.8530	0.1470	1.0582	2.5450
X ₄	4	1	0.1470	0.8530	2.5450	1.0582
X ₅	4	3	0.1468	0.8532	2.5493	1.0574
X ₆	5	2	0.0370	0.9630	3.3448	0.6552

The center vectors of the cluster 1 and cluster 2 are, respectively, (1.6555; 1.9996) and (4.3448; 2.0004).

Figure 2: Membership Degrees and Distances

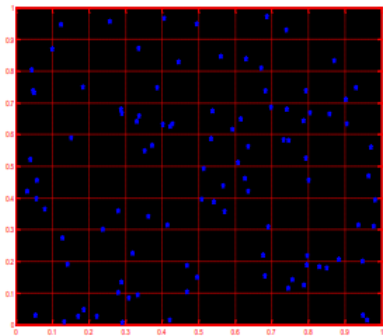


Figure 3: Data Matrix

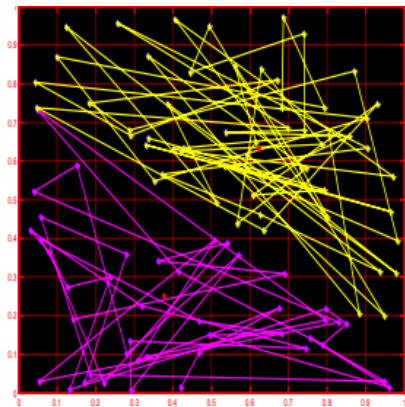


Figure 4: After C means

Fuzzy Transform

Setup:

Let $A_1, A_2, \dots, A_n$ be membership functions on the interval $[a, b]$, and let f be a continuous function on the same interval.

The fuzzy transform of f is given by:

$$F = [F_1, F_2, \dots, F_n]$$

where each F_k is the fuzzy transform component corresponding to the membership function A_k .

Mathematical Formulation:

The k -th component of the fuzzy transform minimizes the following function:

$$L_k = \int_a^b (f(x) - y_k)^2 A_k(x) dx$$

where $A_k(x)$ is the membership function, and y_k represents the fuzzy transform component corresponding to the function f .

Fuzzy Transform

Fuzzy Transform Component F_K :

The k -th fuzzy transform component is calculated using the following formula:

$$F_K = \frac{\int_a^b f(x) A_K(x) dx}{\int_a^b A_K(x) dx}, \quad K = 1, \dots, n$$

Reconstruction of $f(x)$:

Once the components F_K are obtained, the original function $f(x)$ can be reconstructed (approximately) by summing the weighted contributions of the components:

$$f(x) = \sum_{K=1}^n F_K A_K(x)$$

Connection between heuristics and Fuzzy C-Means algorithm

- ▶ The fuzzy c-means algorithm groups elements based on their membership degrees, reflecting similarity to each cluster, displaying closeness to a representative heuristic.
- ▶ The anchoring heuristic adjusts decisions using an initial anchor value to produce the final result.
- ▶ The algorithm starts with a membership degree matrix that is iteratively adjusted to refine the final grouping.

Connection between heuristics and Fuzzy Transform

- ▶ The representativeness heuristic in the Fuzzy Transform involves using weighted mean values, where weights are determined by membership degrees, reflecting the similarity between objects and groups.
- ▶ The anchoring heuristic adjusts decisions based on an initial anchor value to produce the final result in the Fuzzy Transform.
- ▶ The components of the Fuzzy Transform must minimize L_k ; otherwise the components are not valid. Thus, L_k represents an anchor for the Fuzzy Transform.

Conclusion

Applications in cognitive science, ML, stock market analysis, etc.